THE VIRTUAL MEET EXPERIENCE

2021 - 2022

HS VIRTUAL CHALLENGE MEET #3

SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

The Virtual Challenge Meets™

2021-22 Virtual Challenge Meet #3 - Science Test

Biology Questions (B1-B20)

B01. Of the following types of signal molecules below, which one binds to an intracellular receptor and functions as a transcription factor to regulate gene expression?
A) testosterone
B) growth factors
C) acetylcholine
D) bacterial fertility factors

B02. The epithelial ECM is made of extracellular collagen fibers attached to the cell membrane by
A) microfilaments.
B) keratin.
C) proteoglycan.
D) cellulose.

B03. Phylum Chordata are set apart from other animal groups due to synapomorphies such as
A) pharyngeal slits.
B) development of a notochord.
C) hollow nerve cord.
D) only A and B
E) A, B, and C are correct

B04. Interferons are responsible for the heightened immune response to a cell infected with
A) a bacterium.
B) a protist.
C) cancer.
D) a virus.

B05. Transport of materials in a plant must utilize junctions such as plasmodesmata that move materials through the _______ of the cells.
A) symplast
B) ECM
C) apoplast
D) Casparian strip

B06. A species of rye plant creates n=7 gametes. When combined with 2n=14 gametes from a species of wheat the resulting offspring would be considered a(n)
A) diploid.
B) trisomy.
C) haploid.
D) triploid.

B07. Lichens and some microbes can survive in a desiccated state. What type of organism are these classified as?
A) microaerophile
B) xerophile
C) barophile
D) psychrophile

B08. What is an appropriate description of a heterotroph?
A) It fixes atmospheric carbon for organic compounds
B) It releases oxygen as a byproduct
C) It oxidizes organic compounds for energy
D) It breaks water with photon energy

B09. The lac operon of *E. coli* would be considered
A) positive inducible.
B) positive repressible.
C) negative inducible.
D) negative repressible.

B10. How do autopolyploidy and allopolyploidy differ?
A) allopolyploidy involves hybridization from two different species
B) allopolyploidy maintains homologous chromosomes
C) autopolyploidy is caused by mating from different species
D) none of the above are correct

2021-22 Virtual Challenge Meet #3 - Science Test

B11. Individuals with the BRCA-1 gene may develop cancer while others with the gene will not. What occurrence does this describe best?
A) pleiotropy
B) expressivity
C) dominance
D) penetrance

B12. In prokaryotes, the peptidyltransferase activity is found in the _______ ribosomal subunit.
A) 28S
B) 50S
C) 30S
D) 60S
E) 40S

B13. During anoxygenic photosynthesis, water is typically not used as an electron donor; therefore, oxygen is not produced. Which substance below would not be used as an electron donor?
A) H_2
B) CO_2
C) NO_2^-
D) FeS
E) H_2S

B14. The cell lines found from myeloid cells would include all the following, except
A) thrombocytes.
B) erythrocytes.
C) natural killer cells.
D) monocytes.

B15. In fungal cells, the fusion of haploid nuclei to form a diploid nucleus occurs during a process called
A) karyogamy.
B) plasmogamy.
C) mitosis.
D) karyokinesis.

B16. The 3' hydroxyl end of the tRNA is also called the
A) D-loop.
B) anticodon loop.
C) variable loop.
D) acceptor stem.
E) T-loop.

B17. What plant phylum would contain vascularized tissue, pinnate leaves, and spores released by sori structures?
A) Anthophyta
B) Ginkgophyta
C) Chlorophyta
D) Pterophyta

B18. The CDC is currently monitoring an outbreak of _____________ *typhurium* found in pet turtles.
A) *Escherichia*
B) *Clostridium*
C) *Salmonella*
D) *Streptococcus*

B19. ___________ is a process that ensures females, like males, maintain only one functional X-chromosome.
A) Nondisjunction
B) Lyonization
C) Translocation
D) Dicentric chromosome formation

B20. If Gene A and Gene B are found on the same chromosomes and a cross between the following individuals occurs:
AaBb x aabb
With the following offspring counts:

AaBb- 135
Aabb- 54
aaBb- 32
aabb- 119

What is the recombinant frequency?
A) 34%
B) 56%
C) 75%
D) 25%

C01. When an electron of an excited hydrogen atom moves from n = 2 to n = 1, what wavelength of light is emitted?
A) 182 nm
B) 970 nm
C) 480 nm
D) 123 nm
E) 740 nm

C02. An element has the following ground state outer electron configuration (where n represents the highest occupied energy level): $(n-1)d^{10}\ ns^2\ np^4$. Which element could it be?
A) Silicon
B) Sulfur
C) Selenium
D) Germanium
E) none of these

C03. Which of the following has the highest bond order?
A) B_2^-
B) O_2^{2-}
C) O_2^-
D) O_2^{2+}
E) B_2^{2+}

C04. The empirical formula for a compound is found to be NO_2. If 46.0 g of the gas occupies 11.2 L under standard conditions, what is the molar mass of the compound?
A) 46.0
B) 23.0
C) 92.0
D) 30.0
E) 62.0

C05. Consider a 1.0 M solution of an unknown acid, HA. If the pH of the solution is determined to be 4.00, which of the following is true?
A) HA is a strong acid
B) HA is a weak acid with a K_a value around 10^{-4}
C) HA is a weak acid with a K_a value around 10^{-8}
D) HA is a weak acid with a K_a value around 10^{-10}
E) HA would likely be a good acid-base indicator

C06. Methylamine gas, CH_3NH_2, can be classified as
A) a Lewis base only
B) both an Arrhenius and Bronsted-Lowry base
C) a Bronsted-Lowry base only
D) both an Arrhenius and Lewis base
E) both a Bronsted-Lowry and Lewis base

C07. Consider CF_4 and CH_4. Which statement is not correct?
A) Both contain sp^3 hybridized carbon
B) The bond angles in CF_4 are equal to those in CH_4
C) The carbon-fluorine bonds are more polar than the carbon-hydrogen bonds
D) Both are nonpolar molecules
E) All are correct statements.

C08. Which describes the spontaneity of the following reaction?
$NH_4Cl_{(s)} \rightarrow NH_{3(g)} + HCl_{(g)}$ $\Delta H = 188$ kJ
A) Spontaneous at all temperatures
B) Spontaneous only at relatively high temperatures
C) Spontaneous only at relatively low temperatures
D) Not spontaneous at any temperature
E) More information is needed to answer

C09. Which response contains only correct statements?
1. The flow of electrons is spontaneous in voltaic cells
2. In electrolytic cells, electrons flow from the anode to the cathode through the external circuit (wire)
3. The cathode is the positive electrode in voltaic cells

A) 1
B) 3
C) 1 and 2
D) 2 and 3
E) 1, 2 and 3

C10. Azurite, a copper ore, contains 70.0% $Cu_3(CO_3)_2(OH)_2$. What is the percent of copper in the ore?
- A) 18.4 %
- B) 27.6 %
- C) 38.7 %
- D) 55.3 %
- E) 61.2 %

C11. A catalyst
- A) Increases the amount of products present at equilibrium
- B) Increases the rate at which equilibrium is reached but decreases the equilibrium constant
- C) Increases the rate at which equilibrium is reached without changing the equilibrium constant
- D) Increases the ΔH for the process
- E) Lowers the ΔS for the process

C12. What is the enthalpy change in kJ/mol for the complete combustion of C_7H_{16}?
- A) - 3130
- B) - 2930
- C) - 2570
- D) - 2430
- E) + 3130

C13. Ethyl alcohol has a normal boiling point of 78.3 C and ΔH_{vap} = 39.3 kJ/mol.
What is the vapor pressure of ethyl alcohol at 50.0 C?
- A) 118 mmHg
- B) 234 mmHg
- C) 354 mmHg
- D) 485 mmHg
- E) 670. mmHg

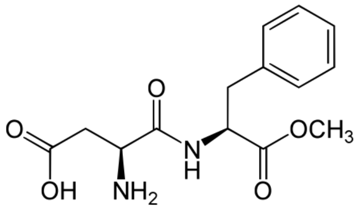
C14. Which of the following would be expected to have the largest radius?
- A) As^{3-}
- B) Br^{-}
- C) Sr^{2+}
- D) Cl^{-}
- E) S^{2-}

C15. According to the kinetic molecular theory of gases
- A) Gaseous particles are in constant, nonlinear motion
- B) At constant temperature, all particles have equal translational kinetic energy
- C) The faster a given particle is moving, the greater its translational kinetic energy
- D) At constant temperature, average kinetic energy is a function of gas volume
- E) All interparticle collisions are completely inelastic

C16. Which statement is not correct?
- A) Activation energy is the same for the forward and reverse reaction
- B) If the forward reaction is endothermic, the reverse will be exothermic
- C) In an endothermic reaction, activation energy is usually greater than the enthalpy
- D) An activated complex has higher energy than any molecule contributing to it
- E) The activated complex will be the highest on the energy profile

C17. What is the minimum pressure that must be applied at 25 C to obtain pure water by reverse osmosis from seawater which is 2.0 M in sodium chloride?
- A) 49 atm
- B) 24 atm
- C) 1.0 atm
- D) 98 atm
- E) 147 atm

C18. Aspartame has the following structure. What is its formula?



A) $C_{10}H_7N_2O_5$
B) $C_{14}H_{12}N_2O_5$
C) $C_{11}H_{17}N_2O_5$
D) $C_{14}H_{18}N_2O_5$
E) $C_{14}H_{17}N_2O_5$

C19. What is the solubility product constant for calcium fluoride in water if its solubility at 25 C is 2.2 x 10^{-4} M?
A) 2.1 x 10^{-14}
B) 4.8 x 10^{-8}
C) 1.1 x 10^{-11}
D) 4.3 x 10^{-11}
E) 2.2 x 10^{-4}

C20. The reaction below is exothermic. Which of the following statements is true?

$$2\ NO_{2\,(g)} \leftrightarrows N_2O_{4\,(g)}$$

A) K_c is negative
B) K_c will decrease as the temperature is increased
C) K_c will not change if the temperature is increased
D) K_c will increase as the temperature is increased
E) None of the statements are correct

P01. According to Michio Kaku, a comet sailed over London in 1682. Edmond Halley traveled to Cambridge to meet with ____________ who showed him that comets move in elliptical paths around the sun and that he could predict their trajectory. On Christmas Day in 1758, the comet sailed over Europe as predicted.
A) Johannes Kepler
B) Galileo Galilei
C) Nicolaus Copernicus
D) Isaac Newton
E) Tycho Brahe

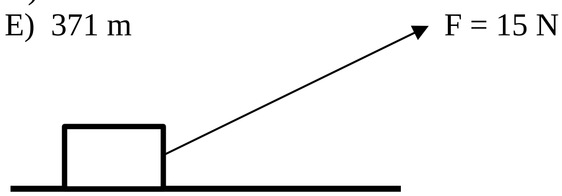
P02. According to Michio Kaku, when some were doubting Einstein's theory of relativity, ___________ wrote to reassure Einstein. He stated, "A new scientific truth does not triumph by convincing its opponents and making them see the light, but rather because the opponents eventually die and a new generation grows up that is familiar with it."
A) Niels Bohr
B) Max Plank
C) Werner Heisenberg
D) Louis de Broglie
E) Erwin Schrodinger

P03. According to Michio Kaku, Joseph von Fraunhofer designed the most precise and accurate spectrograph of his time. He used this device to determine that the sun was mainly made of hydrogen. A new unknown substance was also found. It was named _________, which means "metal from the sun."
A) iron
B) neon
C) helium
D) beryllium
E) lithium

P04. "Above the tropopause and extending upward to an altitude of 50 km above sea level is the ____________."
A) stratosphere
B) troposphere
C) mesosphere
D) thermosphere
E) ionosphere

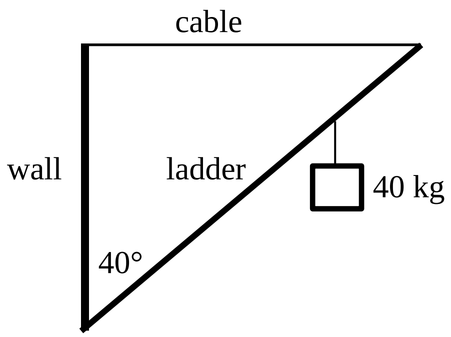
P05. One kilogram is about 2.2 pounds and one inch is about 2.54 centimeters. The density of a substance is 500 lb / ft^3 or about ____ g / cm^3.
A) 4.8
B) 5.6
C) 6.4
D) 7.2
E) 8.0

P06. A plane is flying horizontally at 67.0 m/s, 150 m above the ground. A bowling ball is released from the plane just as the plane flies over a 20.0-m-tall tower. How far from the base of the tower does the bowling ball hit the ground?
A) 327 m
B) 338 m
C) 349 m
D) 360 m
E) 371 m



P07. The coefficient of kinetic friction between the surface of the table and the 3.25-kg block is 0.345. A 15.0-N force pulling force is applied at an angle of 28.8° above the horizontal. Find the acceleration of the block.
A) 1.29 m/s^2
B) 1.43 m/s^2
C) 1.57 m/s^2
D) 1.71 m/s^2
E) 1.85 m/s^2

P08. A 5,000-kg truck was traveling east at a speed of 12.0 m/s when it collided with a 6,000-kg truck traveling south at 10.0 m/s. Immediately after impact, the trucks remained locked together and traveled southeast at a speed of __________ m/s.
A) 7.71
B) 7.91
C) 8.10
D) 8.30
E) 8.49



P09. The foot of a 10.0-m-long ladder rests against a wall. The center of mass of the 12.0-kg ladder is 4.50 m from the base. A 40.0-kg weight hangs from the ladder, 3.00 m from the top. Find the tension in the cable.

A) 261 N
B) 275 N
C) 289 N
D) 303 N
E) 317 N

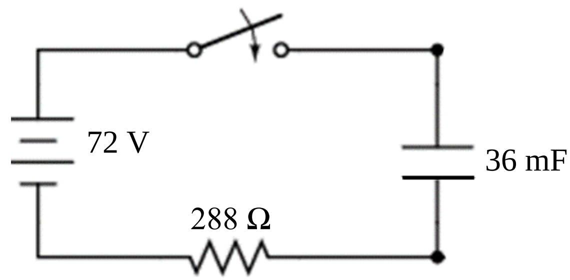
P10. A train traveling 28.5 m/s approaches and then passes a person standing beside the track. The train whistle emits sound with a frequency 1200 Hz. What frequency does the person hear as the train approaches if the temperature is 20.0°C?

A) 1110 Hz
B) 1160 Hz
C) 1210 Hz
D) 1260 Hz
E) 1310 Hz

P11. A 0.842-L glass is filled to the brim with water at T = 12.2°C. How much water will overflow from the glass if the glass and the water are heated to 77.7°C?

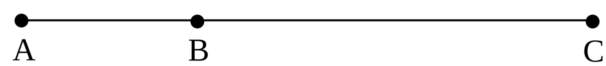
$\left(\beta_G = 9\times10^{-6}\ (\mathrm{C}^\circ)^{-1}, \beta_w = 210\times10^{-6}\ (\mathrm{C}^{\circ-1})\right)$

A) 7.51 mL
B) 8.72 mL
C) 9.93 mL
D) 11.1 mL
E) 12.3 mL



P12. The capacitor is initially uncharged. Find the charge on the capacitor 48.0 ms after the switch is closed.

A) 7.20 mC
B) 8.41 mC
C) 9.62 mC
D) 10.8 mC
E) 12.0 mC



P13. $AB = 2.44\text{ cm},\ BC = 8.62\text{ cm}$

A $+558\ \mu\text{C}$ charge is placed at A and a $-766\ \mu\text{C}$ charge is placed at B. Find the electric field at C due to the charges at A and B.

A) 1.12×10^{4} N/C, left
B) 2.14×10^{5} N/C, left
C) 3.15×10^{6} N/C, left
D) 4.16×10^{7} N/C, left
E) 5.17×10^{8} N/C, left

P14. Two parallel wires each have a length of 24.4 m and each carries a current of 6.48 A dc. The separation of the wires is 8.82 mm. Find the force between the wires.

A) 0.000228 N
B) 0.00230 N
C) 0.0232 N
D) 0.234 N
E) 2.36 N

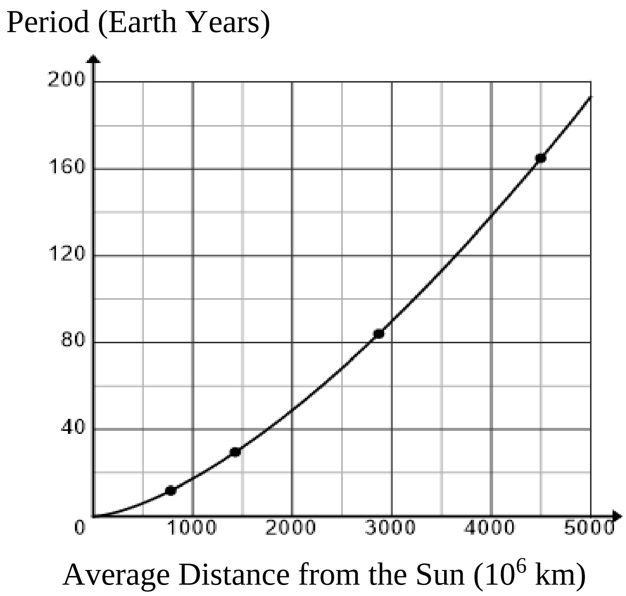
P15. Consider a solenoid that is 48.0-cm long with a cross-sectional area of 12.0 cm^2. It is wound with 240 turns of wire that carry a current of 6.44 A. The relative permeability of the core is 588. Find the magnetic field at a point inside, near the center of the solenoid.
A) 2.38 T
B) 2.57 T
C) 2.76 T
D) 2.95 T
E) 3.14 T

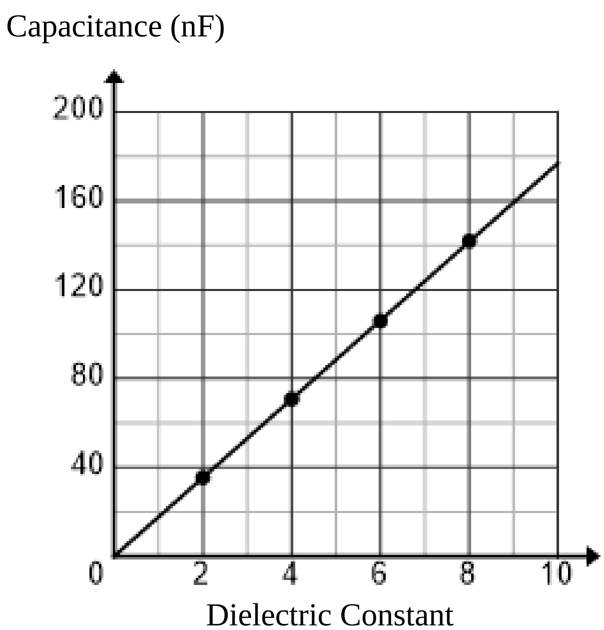
P16. A 2.00-inch-tall candle is placed 24 cm from a convex mirror. The radius of curvature of the mirror is 48 cm. Describe the image formed.
A) virtual, erect, same size
B) virtual, erect, reduced
C) virtual, erect, enlarged
D) virtual, inverted, reduced
E) virtual, inverted, enlarged

P17. Find the kinetic energy of an electron traveling at 0.752c.
A) 0.154 MeV
B) 0.265 MeV
C) 0.376 MeV
D) 0.487 MeV
E) 0.598 MeV

P18. Which of the following particles has a baryon number of $+1$ and a strangeness number of -3?
A) electron
B) proton
C) neutron
D) photon
E) omega

P19. Use the graph on the top of the next column to find the average distance from Earth to the Sun within 1% of the accepted value. The curve is for the period of a body in orbit around the Sun (Kepler's Third Law) and the points plotted are for Mars, Jupiter, Saturn, Uranus and Neptune.
A) 90×10^6 km
B) 120×10^6 km
C) 150×10^6 km
D) 180×10^6 km
E) 210×10^6 km





P20. A parallel-plate capacitor has an area of 2.22 m^2. The capacitance was measured for four different dielectrics that were inserted into and completely filled the gap between the plates. The data was plotted and an LSRL was sketched. Use the graph above to find the separation of the plates within 1% of the true value.
A) 0.00321 mm
B) 0.0445 mm
C) 0.207 mm
D) 1.11 mm
E) 55.9 mm

Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Constants

$R = 0.08206$ L·atm/mol·K
$R = 8.314$ J/mol·K
$R = 62.36$ L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$\mathcal{R} = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg

Water data

$T_{mp} = 0°C$
$T_{bp} = 100°C$
$C_{ice} = 2.09$ J/g K
$C_{water} = 4.184$ J/g K
$C_{steam} = 2.03$ J/g K
$\Delta H_{fus} = 334$ J/g
$\Delta H_{vap} = 2260$ J/g
$K_f = 1.86$ °C/m
$K_b = 0.512$ °C/m

Selected Bond Energies

C - C	348 kJ/mol
C = O	707 kJ/mol
O = O	498 kJ/mol
H - O	464 kJ/mol
C - H	414 kJ/mol

Physics Constants

Free Fall Acceleration	g	9.80 m/s^2
Permittivity of Free Space	ε_0	$8.854\times10^{-12}\ C^2/Nm^2$
Permeability of Free Space	μ_0	$4\pi\times10^{-7}$ Tm/A
Coulomb constant	k	$8.99\times10^{9}\ Nm^2/C^2$
Speed of light in a vacuum	c	3.00×10^{8} m/s
Fundamental charge	e	1.602×10^{-19} C
Planck's constant	h	6.626×10^{-34} Js
Electron mass	m_e	9.11×10^{-31} kg
Proton mass	m_p	1.67265×10^{-27} kg 1.00727647 amu
Neutron mass	m_n	1.67495×10^{-27} kg 1.00866492 amu
Atomic Mass Unit	amu, u	1.66×10^{-27} kg 931 MeV
Gravitational constant	G	$6.67\times10^{-11}\ Nm^2/kg^2$
Stefan-Boltzmann constant	σ	$5.67\times10^{-8}\ W/m^2K^4$
Universal gas constant	R	8.314 J/mol·K
Boltzmann's constant	k_B	1.38×10^{-23} J/K
Electron volts	eV	1.602×10^{-19} J/eV
Rydberg constant	R	$1.097\times10^{7}\ m^{-1}$
Standard Atmospheric Pressure	1 atm	1.013×10^{5} Pa
Horsepower	1 hp	746 W

The Virtual Challenge Meets
Science Contest Answer Sheet

Conference ______ **Grade Level** ______ **Contestant #** ________

Biology	**Chemistry**	**Physics**
B01 ______	C01 ______	P01 ______
B02 ______	C02 ______	P02 ______
B03 ______	C03 ______	P03 ______
B04 ______	C04 ______	P04 ______
B05 ______	C05 ______	P05 ______
B06 ______	C06 ______	P06 ______
B07 ______	C07 ______	P07 ______
B08 ______	C08 ______	P08 ______
B09 ______	C09 ______	P09 ______
B10 ______	C10 ______	P10 ______
B11 ______	C11 ______	P11 ______
B12 ______	C12 ______	P12 ______
B13 ______	C13 ______	P13 ______
B14 ______	C14 ______	P14 ______
B15 ______	C15 ______	P15 ______
B16 ______	C16 ______	P16 ______
B17 ______	C17 ______	P17 ______
B18 ______	C18 ______	P18 ______
B19 ______	C19 ______	P19 ______
B20 ______	C20 ______	P20 ______
B Score	**C Score**	**P Score**

Grader Initials ______ ______ ______

OVERALL SCORE